



Chapter 4

Inverse Manipulator Kinematics

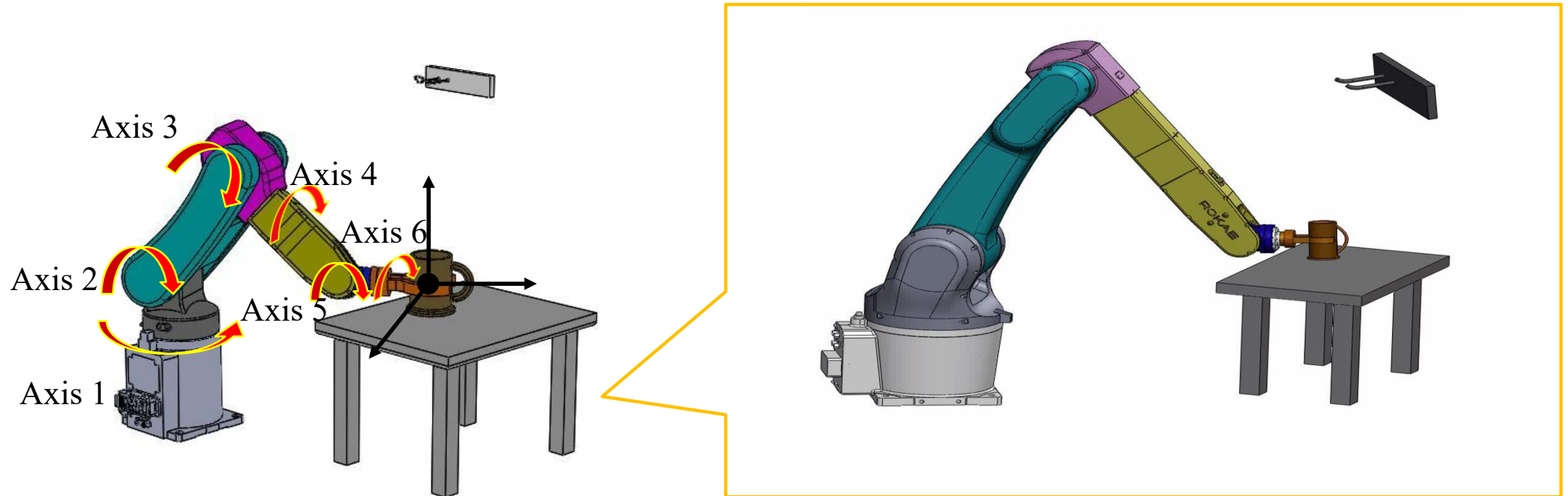
Pengcheng Hu, SMMG, HKUST(GZ)

pengchenghu@hkust-gz.edu.cn

Questions



- ❑ If the position of the end effector of the manipulator in Cartesian space (operator space/task space) is known, how to determine the angular displacements of the joints?
- ❑ Forward Kinematics (FK): joint space $\rightarrow$ Cartesian space (or operation space); Inverse Kinematics (IK): Cartesian space $\rightarrow$ joint space. **Where are they used?**





Chapter 4 Inverse Manipulator Kinematics

- 1. Introduction**
2. Solvability
3. Pieper's solution with three axes intersect
4. The standard frames
5. Examples

1. Introduction

□ Forward Kinematics (FK)

Given θ_i (${}^{i-1}_iT$ can be calculated), find $\{H\}$ or wP

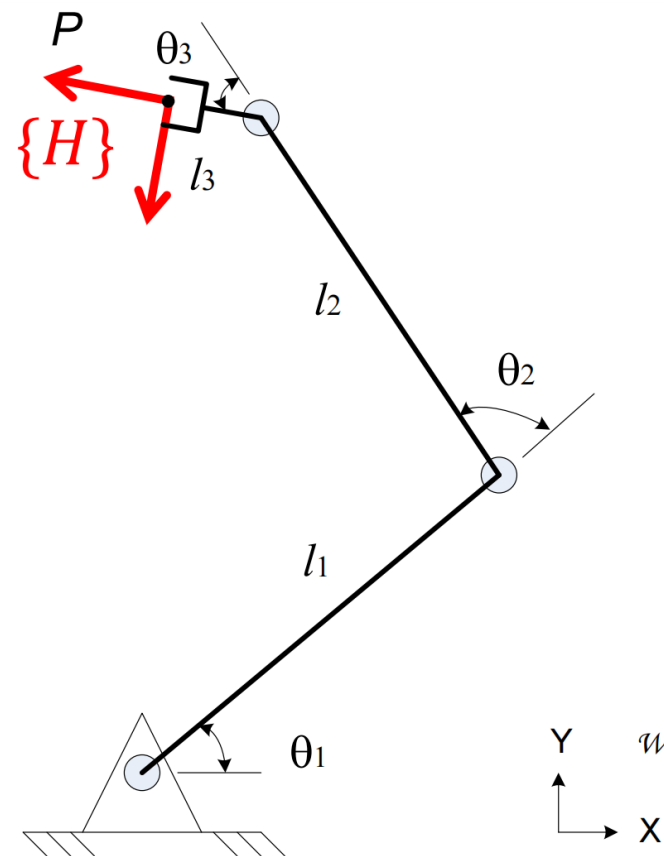
$${}^w_H T = f(\theta_1, \dots, \theta_i, \dots, \theta_n)$$

$${}^wP = {}^w_H T {}^H P$$

□ Inverse Kinematics (IK)

Given $\{H\}$ or wP , find θ_i

$$[\theta_1, \dots, \theta_i, \dots, \theta_n] = f^{-1}({}^w_H T)$$





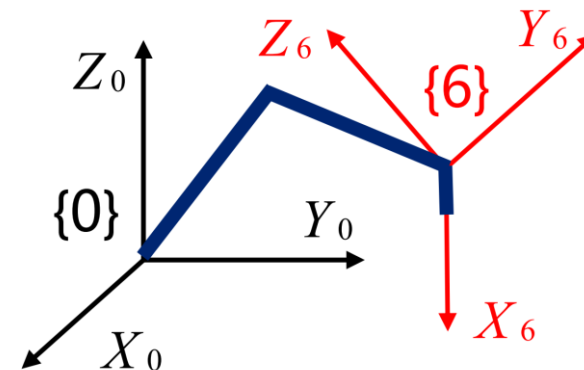
2. Existence of Solutions

- Assume the manipulator has 6 DOFs
 - 6 unknown joint angles (θ_i or d_i , $i = 1, \dots, 6$)
- 0_6T is a 4-by-4 matrix containing 16 numbers, among which 4 are explicitly shown as $[0, 0, 0, 1]$

$${}^0_6T = \begin{bmatrix} \overset{3 \times 3}{{}^0_6R} & \overset{3 \times 1}{{}^0P_{6org}} \\ \hline 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} | & | & | & | \\ {}^0\hat{X}_6 & {}^0\hat{Y}_6 & {}^0\hat{Z}_6 & {}^0P_{6org} \\ | & | & | & | \\ \hline 0 & 0 & 0 & 1 \end{bmatrix}$$

4×4

- Solve for:
 - 12 nonlinear transcendental equations
 - 6 unknowns, 6 constraints (3 normalized constraints, and 3 orthogonal constraints)



2. Existence of Solutions - Workspace

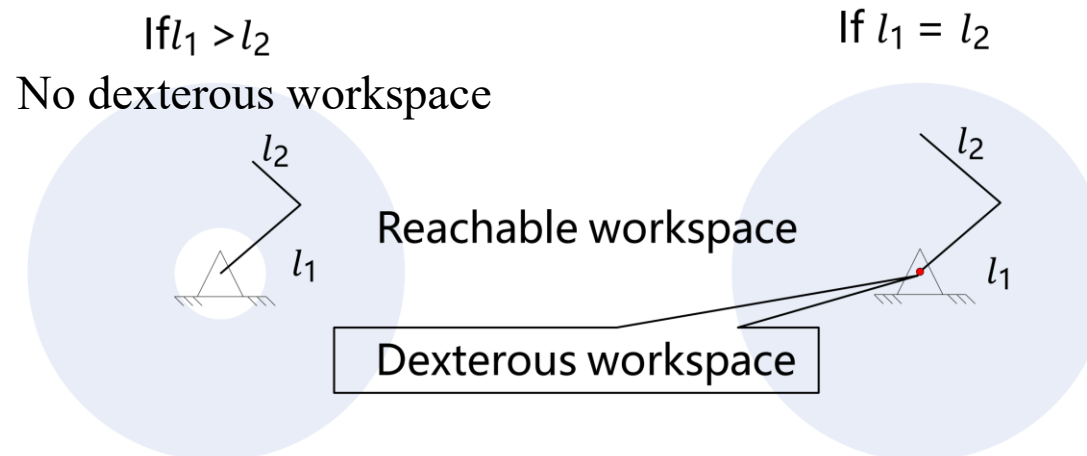
□ Reachable workspace

- Volume of space that the robot can reach in at least one orientation

□ Dexterous workspace

- Volume of space that the robot end-effector can reach with all orientations

Ex: A RR manipulator



2. Existence of Solutions - Workspace

□ Manipulator **subspace** when $n < 6$

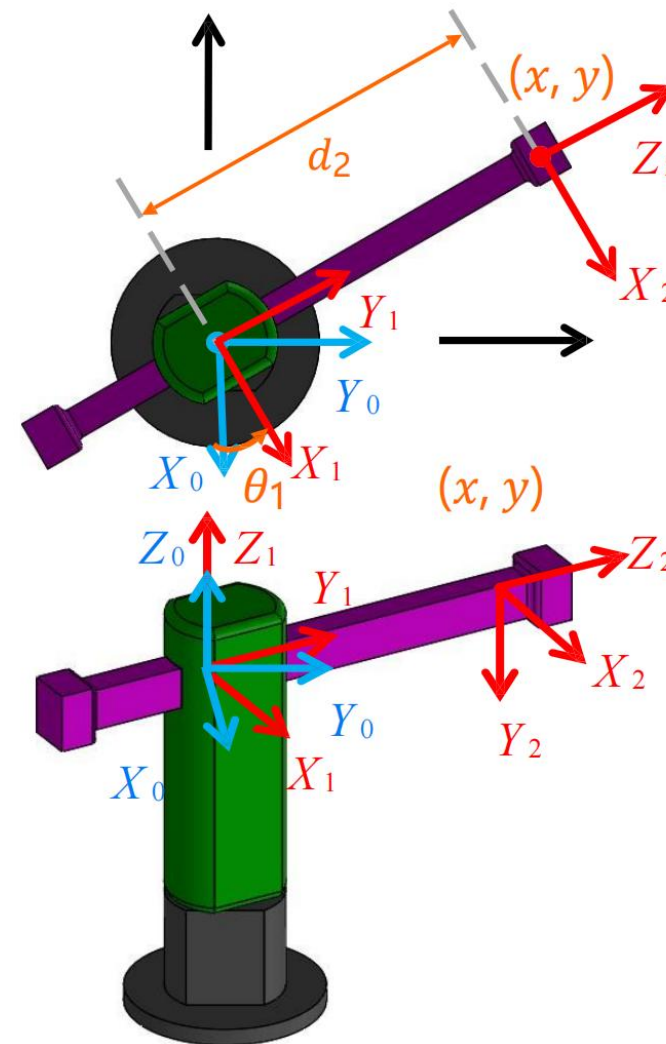
- The space of end effector spanned given arbitrary value of the joints.

Ex: A RP manipulator

- 2-DOF, (x, y) can be arbitrary value and the space thus formed

$${}^w_2T = \begin{bmatrix} \frac{y}{\sqrt{x^2 + y^2}} & 0 & \frac{x}{\sqrt{x^2 + y^2}} & x \\ \frac{-x}{\sqrt{x^2 + y^2}} & 0 & \frac{y}{\sqrt{x^2 + y^2}} & y \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

0Z_2 ${}^0P_{2\text{ ORG}}$



2. Existence of Solutions - Multiple solutions

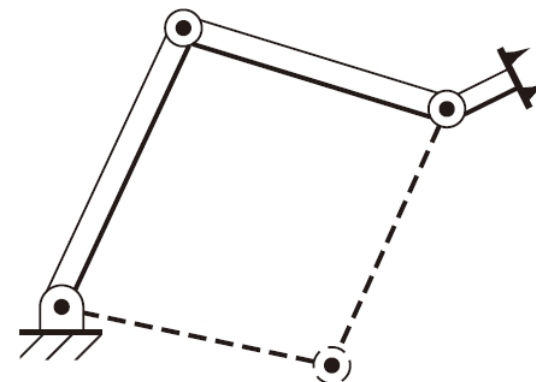
□ Number of solutions

- Since it is nonlinear transcendental equations, 6 unknowns and 6 equations does not guarantee a unique solution.

□ Determined by the number of joints and the link parameter

- Ex: A RRRRRR manipulator, for different link length:

a_i	Number of solutions
$a_1 = a_3 = a_5 = 0$	≤ 4
$a_3 = a_5 = 0$	≤ 8
$a_3 = 0$	≤ 16
All $a_i \neq 0$	≤ 16



Three-link manipulator - dashed lines indicate a second solution

2. Existence of Solutions - Multiple solutions

□ Ex: PUMA560 (6 rotational joints)

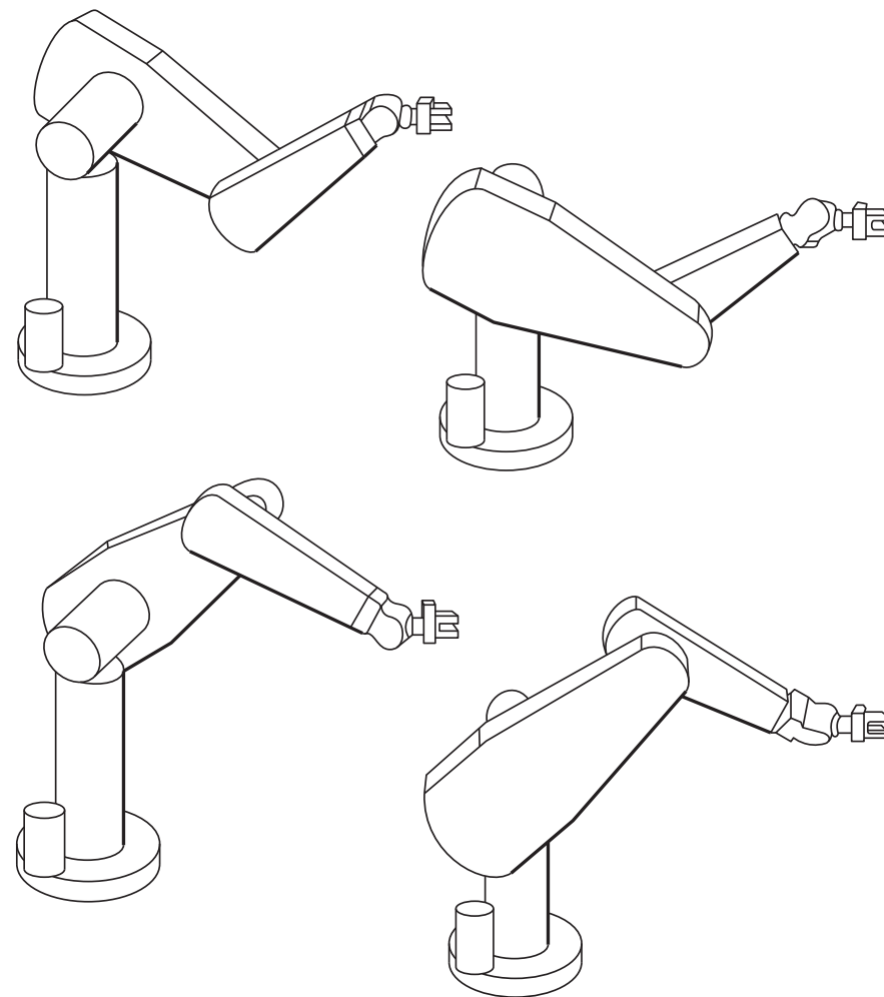
- For specific points in the working space, 8 groups of solutions
- The first 3 axes have 4 postures, shown in the figure to the right
- 2 sets of wrist rotation postures in each posture

$$\theta'_4 = \theta_4 + 180^\circ$$

$$\theta'_5 = -\theta_5$$

$$\theta'_6 = \theta_6 + 180^\circ$$

- If there are geometric constraints for the manipulator (e.g., constraints on the joints), some postures are infeasible

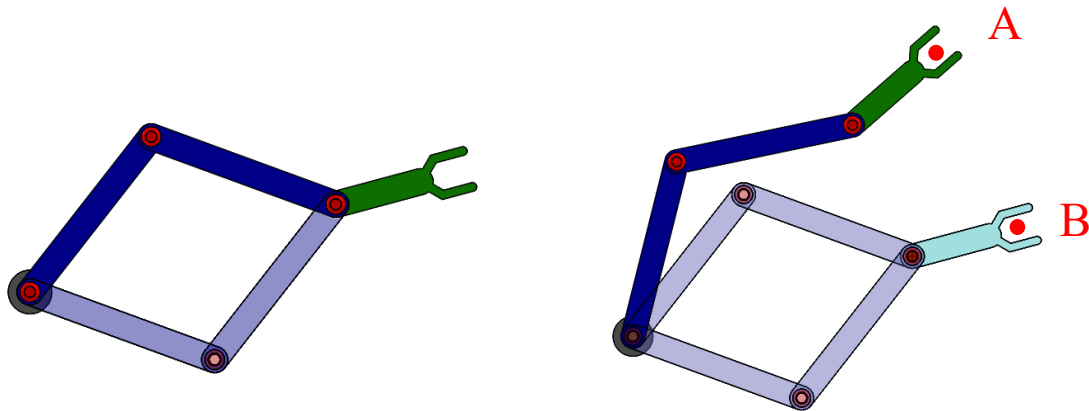


Four solutions of the PUMA 560

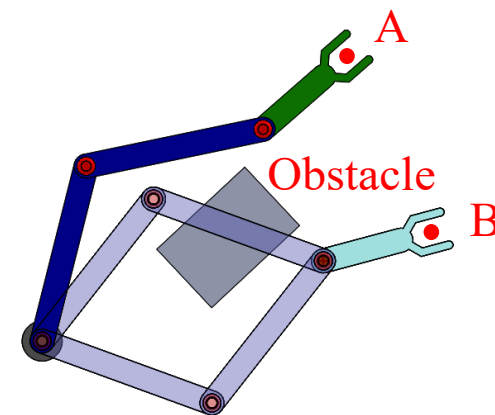
2. Existence of Solutions - Multiple solutions

□ If there are multiple solutions, the way the solutions are chosen

- The closest solution to the current state
- Minimal energy consumption
- Avoiding obstacles
- ...



Moving from A to B, multiple solutions



Obstacle avoidance, only one solution



2. Existence of Solutions - Solution Methods

□ Solution methods of IK:

- Closed-form solutions: **algebraic** or **geometric** methods
- Numerical solutions: iteration-based methods (popular in research paper)
- Machine learning based methods: Physics-informed machine learning (hot research direction)
- Currently, most robotic arms are designed to have analytical solutions: **Pieper's solution**
(intersection of three neighboring axes)

2. Existence of Solutions - Ex1: RRR manipulator

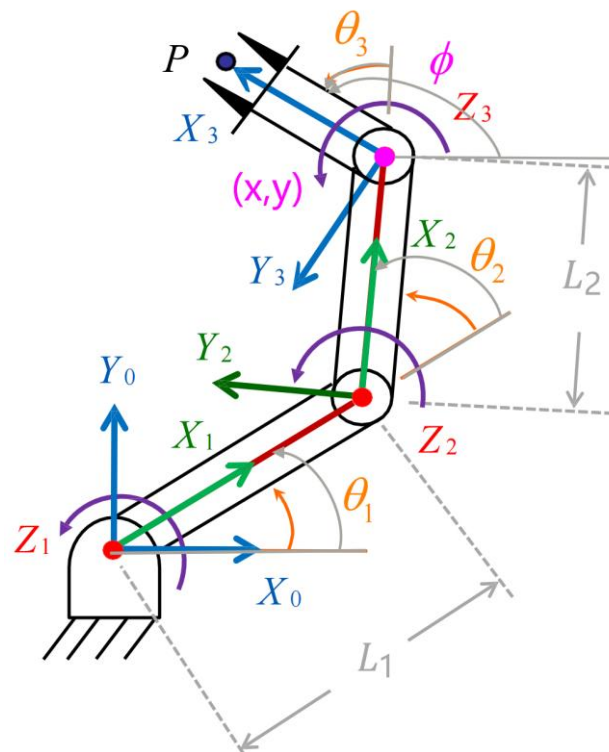
□ IK problem: given (x, y, ϕ) , calculate $(\theta_1, \theta_2, \theta_3) = ?$

● Forward kinematics

$${}^0_3T = \begin{bmatrix} c_{123} & -s_{123} & 0.0 & l_1c_1 + l_2c_{12} \\ s_{123} & c_{123} & 0.0 & l_1s_1 + l_2s_{12} \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Goal point: a specified wrist frame relative to the base frame

$${}^0_3T = \begin{bmatrix} c_\phi & -s_\phi & 0.0 & x \\ s_\phi & c_\phi & 0.0 & y \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$





2. Existence of Solutions - Ex1: RRR manipulator

□ **Geometric method:** based on the planar geometry

Solve θ_2

$$x^2 + y^2 = l_1^2 + l_2^2 - 2l_1l_2 \cos(180^\circ - \theta_2)$$

$$c_2 = \frac{x^2 + y^2 - l_1^2 - l_2^2}{2l_1l_2}$$

Solve θ_1

Find β , and $\beta = \text{Atan2}(y, x)$

According to the **law of Cosines** $\cos \psi = \frac{l_2^2 - (x^2 + y^2) - l_1^2}{-2l_1\sqrt{x^2 + y^2}}$ $0^\circ < \psi < 180^\circ$

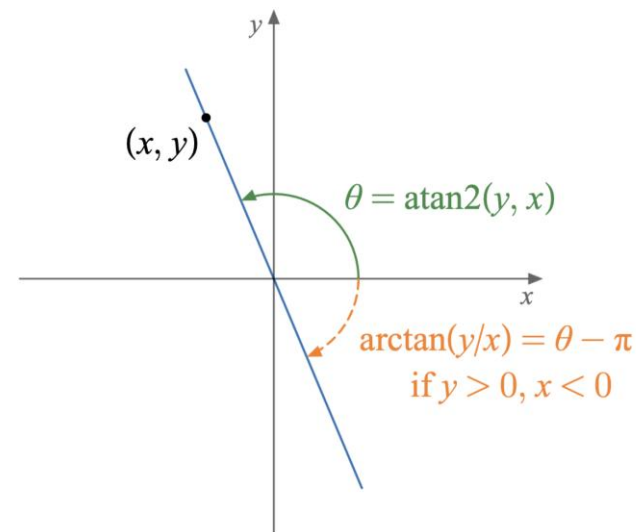
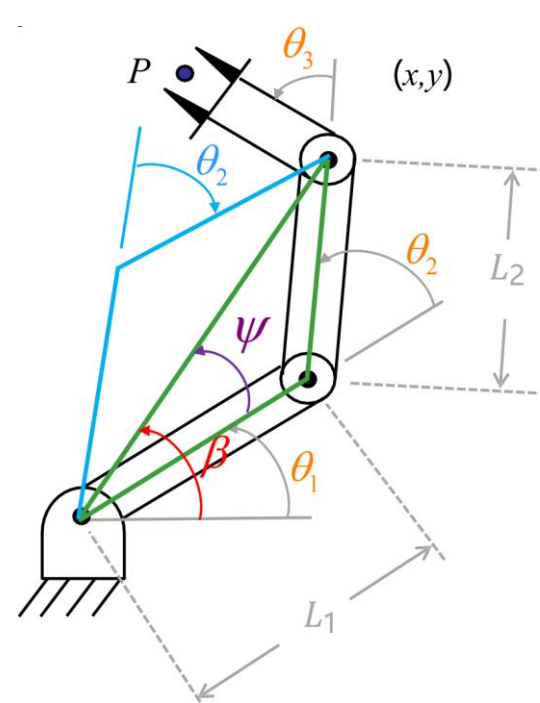
According to the figure to the right

$$\text{Finally, it has } \theta_1 = \begin{cases} \text{atan2}(y, x) + \psi & \theta_2 < 0^\circ \\ \text{atan2}(y, x) - \psi & \theta_2 > 0^\circ \end{cases}$$

Solve θ_3

The three angles is summed to compute the orientation of the last link, i.e.,

$$\theta_3 = \phi - \theta_1 - \theta_2$$



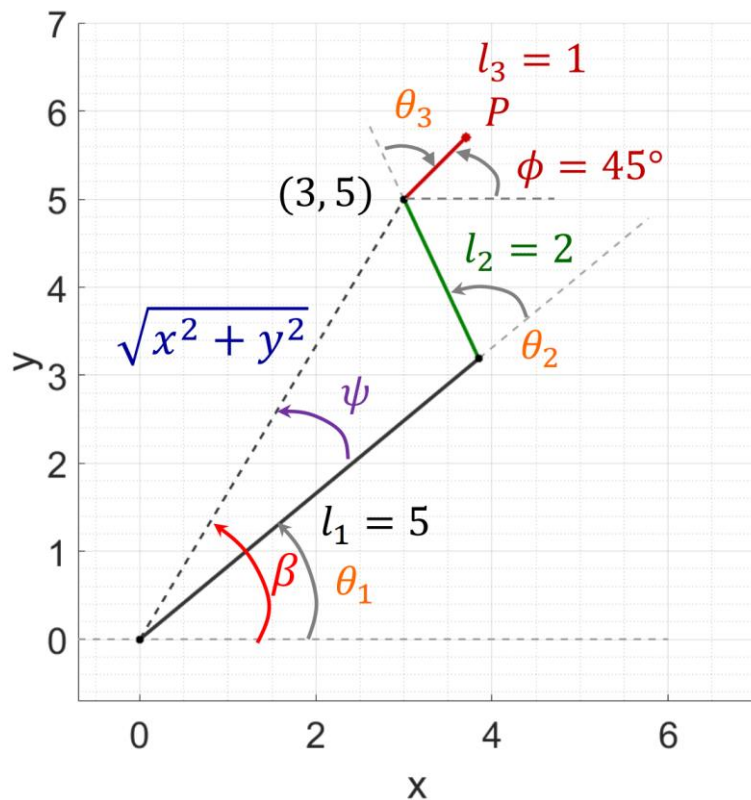
Function of $\text{atan2}(y, x)$



2. Existence of Solutions - Ex1: RRR manipulator

□ Calculation toward the specific example

$$(x, y, \phi) = (3, 5, 45^\circ)$$



$$c_2 = \frac{x^2 + y^2 - l_1^2 - l_2^2}{2l_1l_2}$$

$$\theta_2 = 75.5^\circ$$

$$\cos\psi = \frac{l_2^2 - (x^2 + y^2) - l_1^2}{-2l_1\sqrt{x^2 + y^2}}$$

$$\psi = 19.4^\circ$$

$$\theta_1 = \text{atan2}(y, x) - \psi$$

$$\theta_1 = 39.6^\circ$$

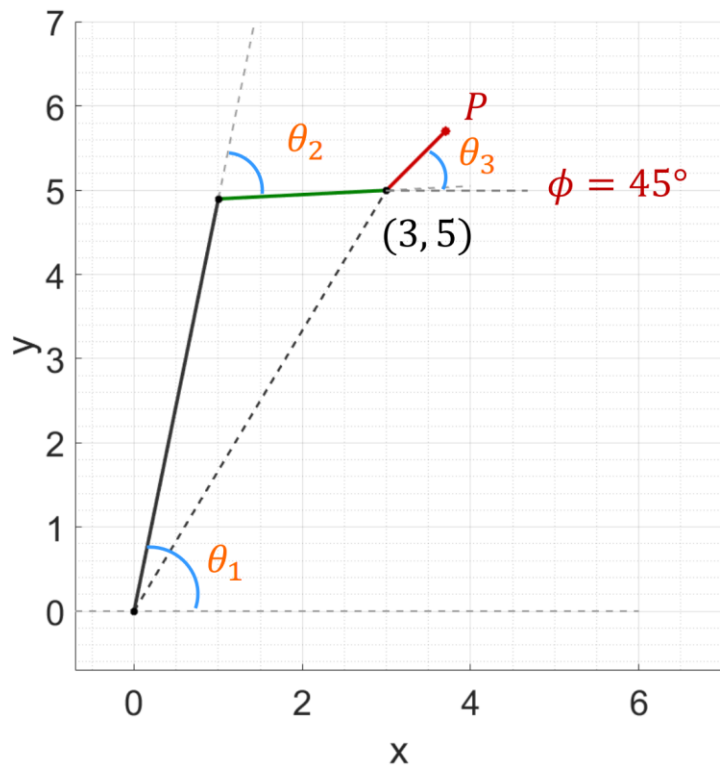
$$\theta_3 = \phi - \theta_1 - \theta_2$$

$$\theta_3 = -70.2^\circ$$



2. Existence of Solutions - Ex1: RRR manipulator

□ **Assignment:** For the same displacement and posture, solve for another set $(\theta_1, \theta_2, \theta_3)$.



(A)
 $\theta_1 = 75.5$
 $\theta_2 = -78.4$
 $\theta_3 = 42.1$

(B)
 $\theta_1 = 78.4$
 $\theta_2 = -75.5$
 $\theta_3 = 42.1$

(C)
 $\theta_1 = -78.4$
 $\theta_2 = 75.5$
 $\theta_3 = 42.1$

(D)
 $\theta_1 = 59$
 $\theta_2 = -75.5$
 $\theta_3 = 42.1$



2. Existence of Solutions - Ex1: RRR manipulator

□ Algebraic solution

- Establishing the Equation

$$c_\phi = c_{123}$$

$$s_\phi = s_{123}$$

$$x = l_1 c_1 + l_2 c_{12}$$

$$y = l_1 s_1 + l_2 s_{12}$$

$${}^0_3T = \begin{bmatrix} c_{123} & -s_{123} & 0.0 & l_1 c_1 + l_2 c_{12} \\ s_{123} & c_{123} & 0.0 & l_1 s_1 + l_2 s_{12} \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} c_\phi & -s_\phi & 0.0 & x \\ s_\phi & c_\phi & 0.0 & y \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Solve θ_2

$$x^2 + y^2 = l_1^2 + l_2^2 + 2l_1 l_2 c_2$$

$$c_2 = \frac{x^2 + y^2 - l_1^2 - l_2^2}{2l_1 l_2}$$

> 1 or < -1: too far for the manipulator to reach

$-1 \leq c_2 \leq 1$: "two solutions" $s_2 = \pm \sqrt{1 - c_2^2}$.

And, $\theta_2 = \text{Atan2}(s_2, c_2)$



2. Existence of Solutions - Ex1: RRR manipulator

- Bring θ_2 into the equation, and write x and y as a function of θ_1

$$x = l_1 c_1 + l_2 c_{12} = (l_1 + l_2 c_2) c_1 + (-l_2 s_2) s_1 \triangleq k_1 c_1 - k_2 s_1$$

$$y = l_1 s_1 + l_2 s_{12} = (l_1 + l_2 c_2) s_1 + (l_2 s_2) c_1 \triangleq k_1 s_1 + k_2 c_1$$

where,

$$k_1 = l_1 + l_2 c_2,$$
$$k_2 = l_2 s_2.$$

- Change of variables

define

$$r = +\sqrt{k_1^2 + k_2^2}$$

$$\gamma = \text{Atan2}(k_2, k_1)$$

then

$$k_1 = r \cos \gamma$$

$$k_2 = r \sin \gamma$$

And then

$$\frac{x}{r} = \cos \gamma \cos \theta_1 - \sin \gamma \sin \theta_1 = \cos(\gamma + \theta_1)$$

$$\frac{y}{r} = \cos \gamma \sin \theta_1 + \sin \gamma \cos \theta_1 = \sin(\gamma + \theta_1)$$

- Solve θ_1

$$\gamma + \theta_1 = \text{Atan2}\left(\frac{y}{r}, \frac{x}{r}\right) = \text{Atan2}(y, x)$$

$$\Rightarrow \theta_1 = \text{Atan2}(y, x) - \text{Atan2}(k_2, k_1)$$

Choose a different sign (+ or -) for θ_2 , c_2 and s_2 change, k_1 and k_2 then change, and θ_1 eventually changes

2. Existence of Solutions - Ex1: RRR manipulator

- The Equation

$$c_\phi = c_{123}$$

$$s_\phi = s_{123}$$

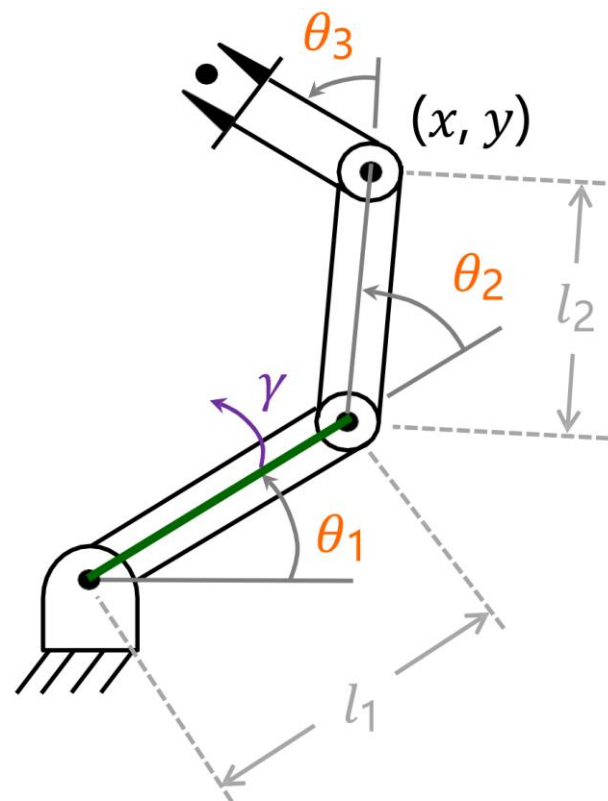
$$x = l_1 c_1 + l_2 c_{12}$$

$$y = l_1 s_1 + l_2 s_{12}$$

- Solve θ_3

$$\theta_1 + \theta_2 + \theta_3 = \text{Atan2}(s_\phi, c_\phi) = \phi$$

$$\Rightarrow \theta_3 = \phi - \theta_1 - \theta_2$$





2. Existence of solutions – Algebraic solution by reduction to polynomial

□ **Ex:** How to solve the θ of $a\cos\theta + b\sin\theta = c$?

- Method: transform to polynomials (with analytic solution up to order 4)

$$\tan\left(\frac{\theta}{2}\right) = u, \quad \cos\theta = \frac{1-u^2}{1+u^2}, \quad \sin\theta = \frac{2u}{1+u^2}$$

- Steps:

$$a\cos\theta + b\sin\theta = c$$

$$a\frac{1-u^2}{1+u^2} + b\frac{2u}{1+u^2} = c$$

$$(a+c)u^2 - 2bu + (c-a) = 0$$

$$u = \frac{b \pm \sqrt{b^2 + a^2 - c^2}}{a+c}$$

$$\theta = 2 \tan^{-1} \left(\frac{b \pm \sqrt{b^2 + a^2 - c^2}}{a+c} \right) \quad a+c \neq 0$$

$$\theta = 180^\circ \quad a+c = 0$$

There is a limit to the value of a, b, c , does not guarantee a solution

3. Pieper's Solution for Triaxial Intersection

□ A general robot with six degrees of freedom does not have a closed-form solution, certain special cases can be solved.

□ **Pieper's law: If the 6-DOF manipulator has three consecutive axes intersecting at the same point, then the manipulator has an analytic IK solution.**

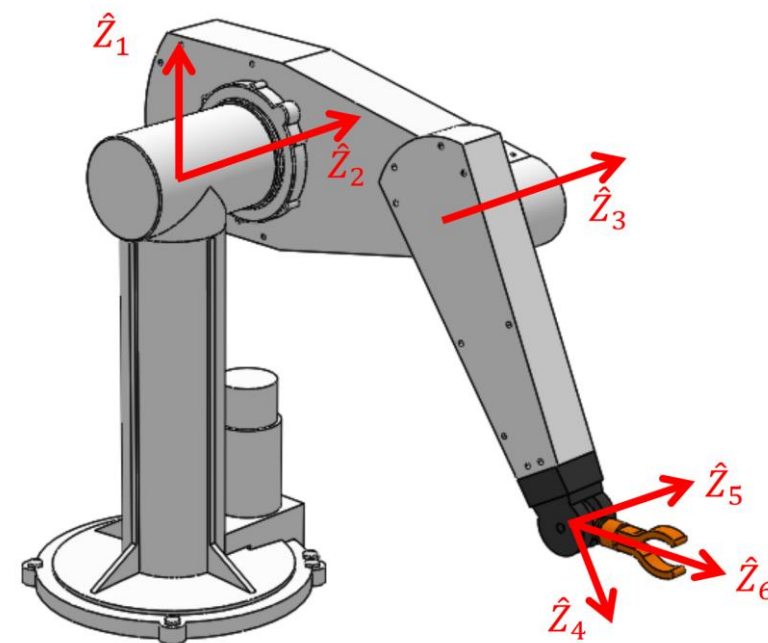
□ Typically, the manipulator is designed as follows:

- First three axes: generating movement (position)
- Last three axes: generating rotation (posture)

□ Ex: A RRRRRR manipulator

Since the rear three axes intersect at a point, it has

$${}^0P_{6\text{ org}} = {}^0P_{5\text{ org}} = {}^0P_{4\text{ org}}$$



PUMA 560

Origins of link frames {4}, {5}, and {6} are all located at this point of intersection



3. Pieper's Solution for Triaxial Intersection

□ Positioning structure

- Rule: Isolate $\theta_1, \theta_2, \theta_3$

The intersection point in base frame

$$\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = {}^0P_{4ORG} = {}^0T_1 {}^1T_2 {}^2T_3 {}^3P_{4ORG}$$

Note: ${}^{i-1}T_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$$= {}^0T_1 {}^1T_2 {}^2T_3 \begin{bmatrix} a_3 \\ -d_4 s\alpha_3 \\ d_4 c\alpha_3 \\ 1 \end{bmatrix} = {}^0T_1 {}^1T_2 \begin{bmatrix} f_1(\theta_3) \\ f_2(\theta_3) \\ f_3(\theta_3) \\ 1 \end{bmatrix}$$

4^{th} column of 3T_4

$\cos \theta_i = c\theta_i = c_i$
 $\sin \theta_i = s\theta_i = s_i$

And,

$$\begin{bmatrix} f_1(\theta_3) \\ f_2(\theta_3) \\ f_3(\theta_3) \\ 1 \end{bmatrix} = {}^2T_3 \begin{bmatrix} a_3 \\ -d_4 s\alpha_3 \\ d_4 c\alpha_3 \\ 1 \end{bmatrix}$$

Separate $\theta_1, \theta_2, \theta_3$, f is a function of θ_3

$$f_1(\theta_3) = a_3 c_3 + d_4 s\alpha_3 s_3 + a_2$$

$$f_2(\theta_3) = a_3 c\alpha_2 s_3 - d_4 s\alpha_3 c\alpha_2 c_3 - d_4 s\alpha_2 c\alpha_3 - d_3 s\alpha_2$$

$$f_3(\theta_3) = a_3 s\alpha_2 s_3 - d_4 s\alpha_3 s\alpha_2 c_3 + d_4 c\alpha_2 c\alpha_3 + d_3 c\alpha_2$$



3. Pieper's Solution for Triaxial Intersection

- Next step

$${}^0P_{4ORG} = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = {}^0T_1 {}^1T_2 \begin{bmatrix} f_1(\theta_3) \\ f_2(\theta_3) \\ f_3(\theta_3) \\ 1 \end{bmatrix} = {}^0T_1 \begin{bmatrix} g_1(\theta_2, \theta_3) \\ g_2(\theta_2, \theta_3) \\ g_3(\theta_2, \theta_3) \\ 1 \end{bmatrix} = \begin{bmatrix} c_1 g_1 - s_1 g_2 \\ s_1 g_1 + c_1 g_2 \\ g_3 \\ 1 \end{bmatrix}$$

Separate $\theta_1, \theta_2, \theta_3$, g is a function of θ_2 and θ_3

$$g_1(\theta_2, \theta_3) = c_2 f_1 - s_2 f_2 + a_1$$

$$g_2(\theta_2, \theta_3) = s_2 c \alpha_1 f_1 + c_2 c \alpha_1 f_2 - s \alpha_1 f_3 - d_2 s \alpha_1$$

$$g_3(\theta_2, \theta_3) = s_2 s \alpha_1 f_1 + c_2 s \alpha_1 f_2 + c \alpha_1 f_3 + d_2 c \alpha_1$$

Write an expression for the squared magnitude of ${}^0P_{4ORG}$

$$\begin{aligned} r &= x^2 + y^2 + z^2 = g_1^2 + g_2^2 + g_3^2 \\ &= f_1^2 + f_2^2 + f_3^2 + a_1^2 + d_2^2 + 2d_2 f_3 + 2a_1(c_2 f_1 - s_2 f_2) \\ &= (k_1 c_2 + k_2 s_2) 2a_1 + k_3 \end{aligned}$$

$$k_1(\theta_3) = f_1$$

$$k_2(\theta_3) = -f_2$$

$$k_3(\theta_3) = f_1^2 + f_2^2 + f_3^2 + a_1^2 + d_2^2 + 2d_2 f_3$$

r is only function of θ_2 and θ_3



3. Pieper's Solution for Triaxial Intersection

- Besides

$$z = g_3 = (k_1 s_2 - k_2 c_2) s \alpha_1 + k_4 \quad \text{and} \quad k_4(\theta_3) = f_3 c \alpha_1 + d_2 c \alpha_1$$

z is only function of θ_2 and θ_3

- Integrate r and z together for consideration

$$\begin{cases} r = (k_1 c_2 + k_2 s_2) 2a_1 + k_3 \\ z = (k_1 s_2 - k_2 c_2) s \alpha_1 + k_4 \end{cases} \quad \boxed{\text{dependence on } \theta_1 \text{ has been eliminated}}$$

$$\begin{aligned} \text{If } a_1 = 0, r &= k_3(\theta_3) = f_1^2 + f_2^2 + f_3^2 + a_1^2 + d_2^2 + 2d_2 f_3 \\ \text{If } s \alpha_1 = 0, z &= k_4(\theta_3) = f_3 c \alpha_1 + d_2 c \alpha_1 \end{aligned} \quad \left. \vphantom{\begin{aligned} \text{If } a_1 = 0, r &= k_3(\theta_3) = f_1^2 + f_2^2 + f_3^2 + a_1^2 + d_2^2 + 2d_2 f_3 \\ \text{If } s \alpha_1 = 0, z &= k_4(\theta_3) = f_3 c \alpha_1 + d_2 c \alpha_1 \end{aligned}} \right\} \boxed{\text{Function of } \theta_3 \text{ only}}$$

$$\text{Else } \frac{(r - k_3)^2}{4a_1^2} + \frac{(z - k_4)^2}{s^2 \alpha_1} = k_1^2 + k_2^2 \quad \boxed{\text{Eliminate } s_2 \text{ and } c_2}$$

Solve an equation of degree 4 to get θ_3 of all three cases by using " $u = \tan(\frac{\theta_3}{2})$ "

3. Pieper's Solution for Triaxial Intersection

□ Eventually

Using $\begin{cases} r = (k_1 c_2 + k_2 s_2) 2a_1 + k_3 \\ z = (k_1 s_2 - k_2 c_2) s\alpha_1 + k_4 \end{cases}$ to solve θ_2

Using $\begin{cases} x = c_1 g_1(\theta_2, \theta_3) - s_1 g_2(\theta_2, \theta_3) \\ y = s_1 g_1(\theta_2, \theta_3) + c_1 g_2(\theta_2, \theta_3) \end{cases}$ to solve θ_1

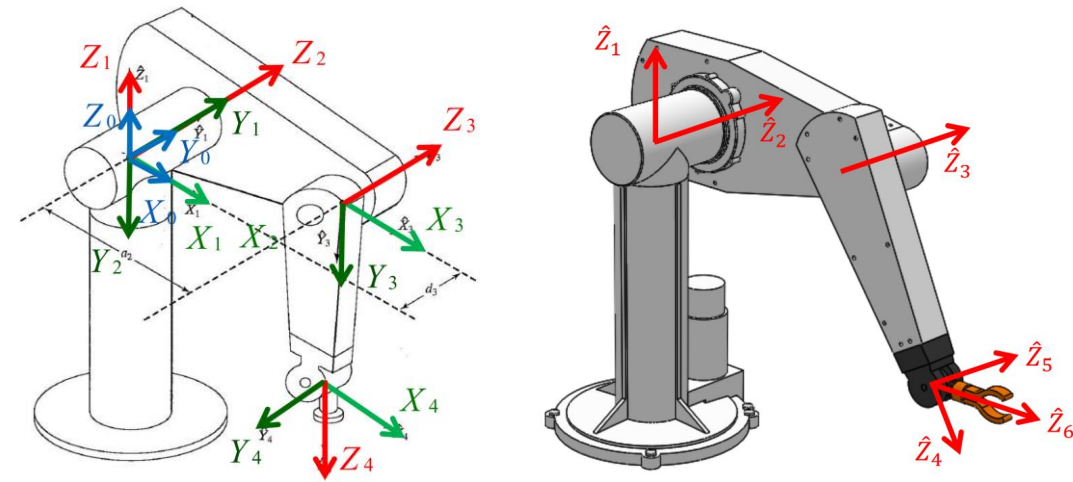
□ Orientation

- Given $\theta_1, \theta_2, \theta_3$

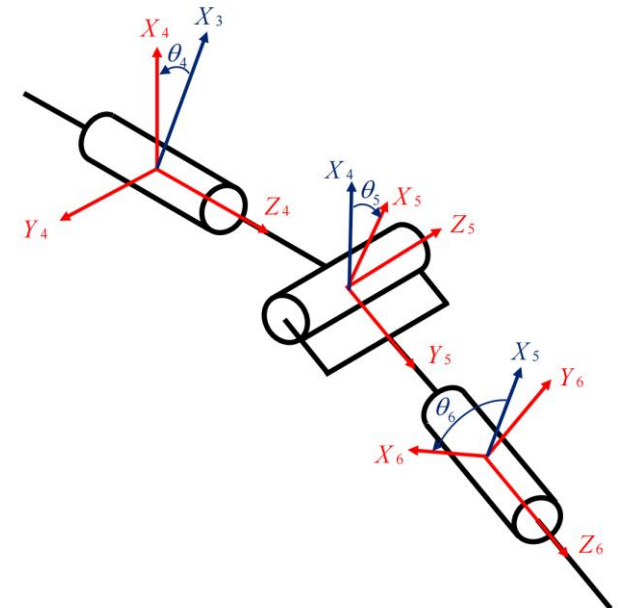
$${}^3_6R = {}^0_3R^{-1} {}^0_6R$$

- Solving for $\theta_4, \theta_5, \theta_6$ in terms of Z-Y-Z Euler angles

$${}^3_6R = {}^3_4R {}^4_6R = {}^3_4R(\theta_4 = 0) R_{ZYZ}(\alpha, \beta, \gamma)$$



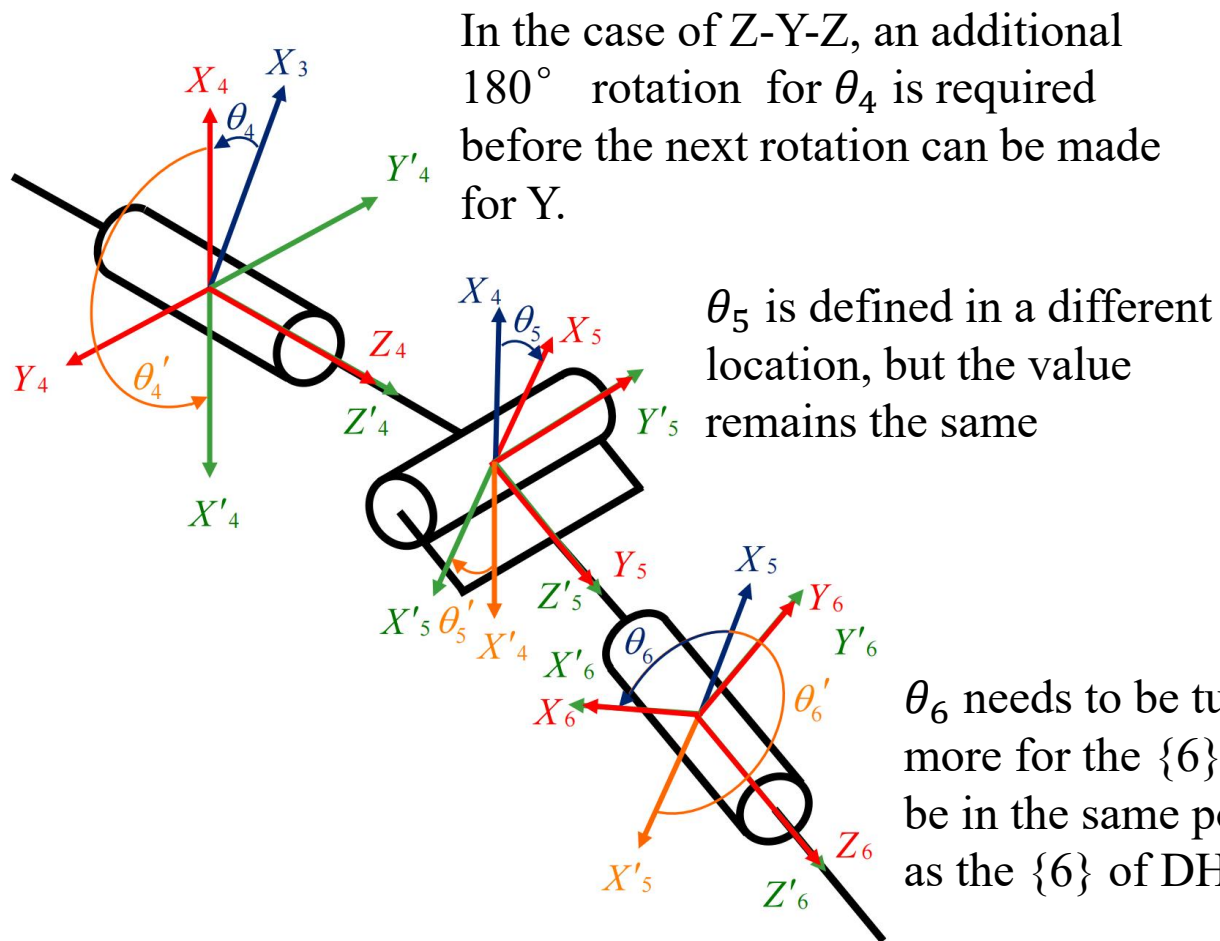
DH definition of PUMA 560



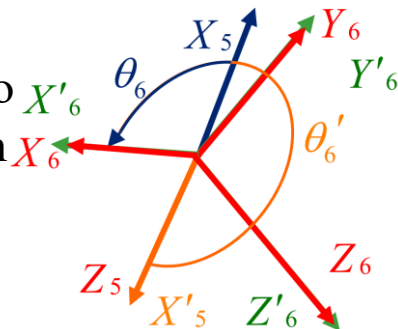
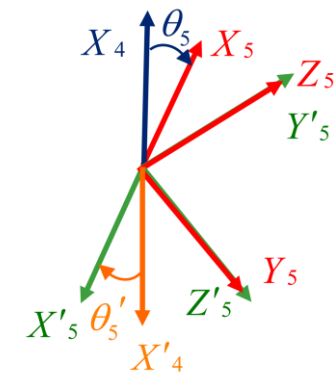
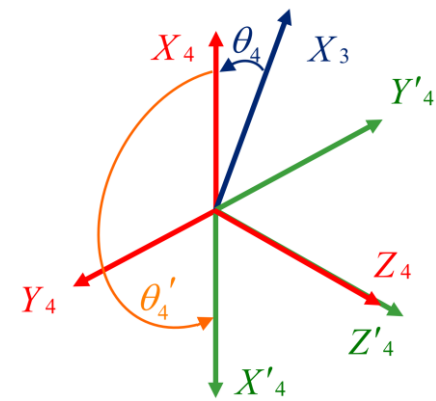
Joints 4-6

3. Pieper's Solution for Triaxial Intersection

□ DH definition vs. Z-Y-Z Euler Angles

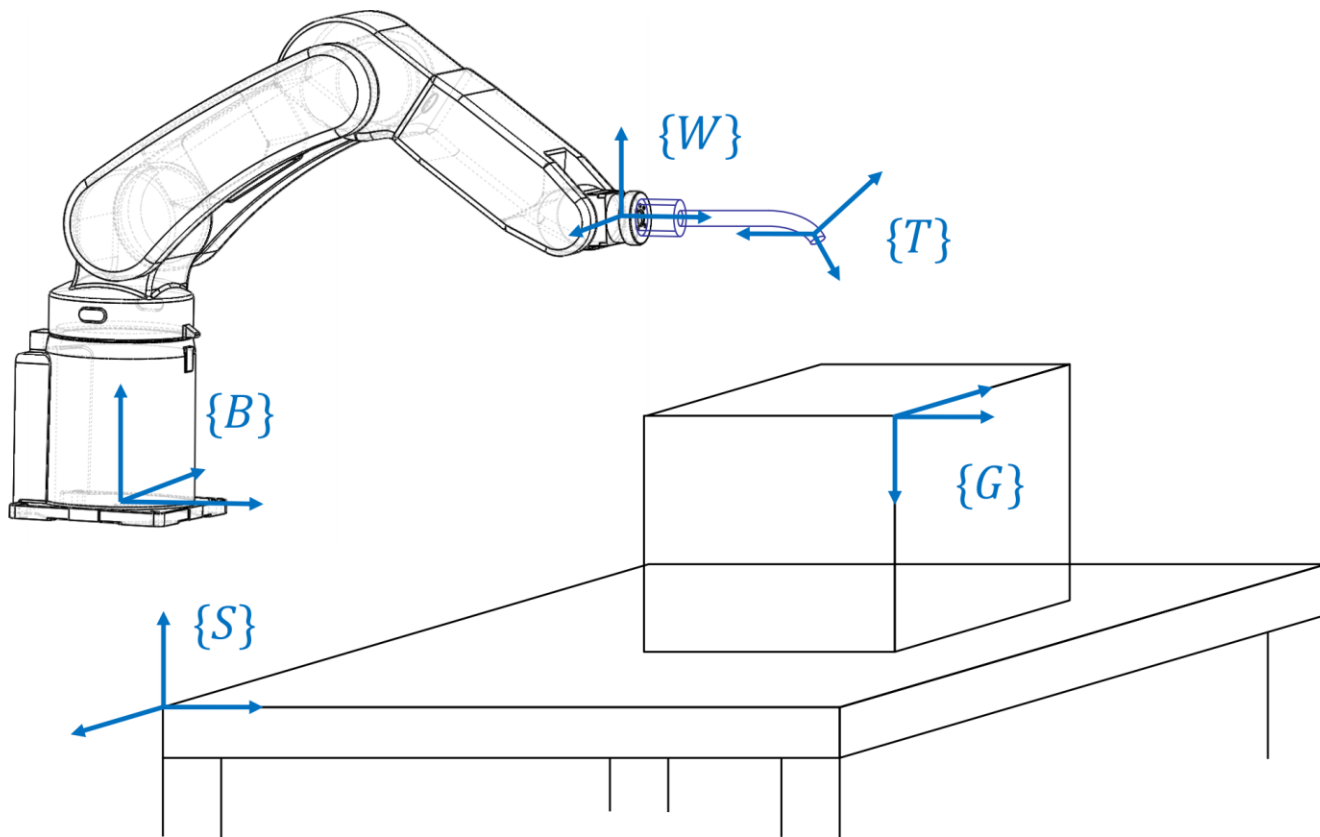


θ_6 needs to be turned 180° more for the $\{6\}$ of Z-Y-Z to be in the same posture again as the $\{6\}$ of DH



4. The standard frames

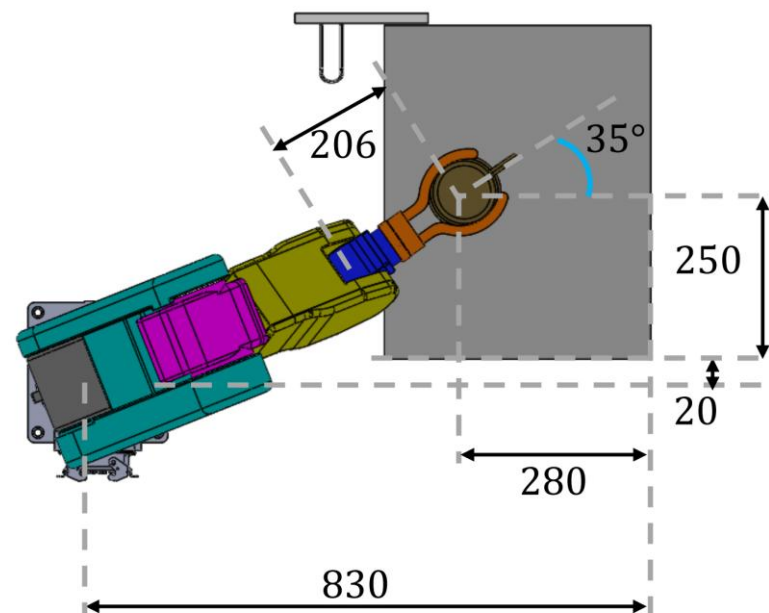
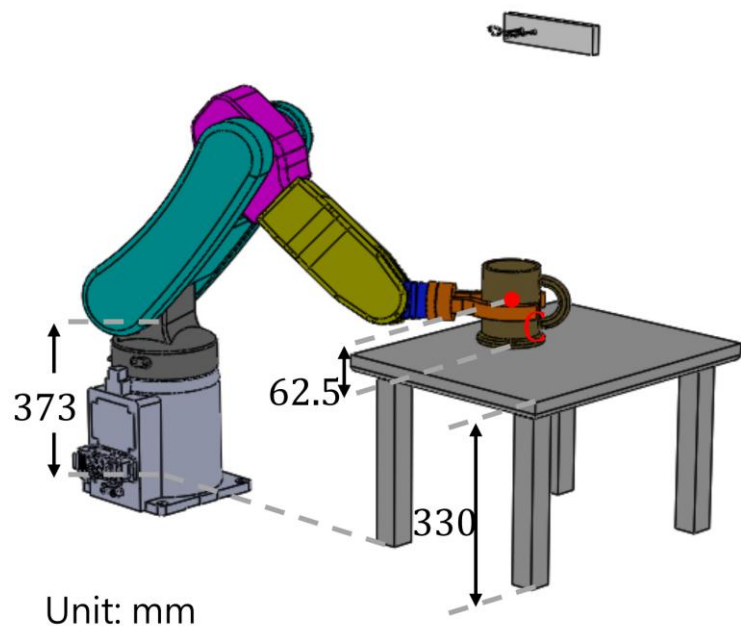
□ Base, wrist, tool, station, and goal frames



- The station frame $\{S\}$ is defined relative to the base frame $\{B\}$.
- The user specifies the description of the tool being used by the robot by giving the $\{T\}$ -frame specification, i.e., $\{T\}$ is specified relative to $\{W\}$.
- The user specifies the goal point for a robot motion by giving the description of the goal frame, $\{G\}$, relative to the station frame $\{S\}$.
- The robot system calculates a series of joint angles to move the joints through in order that the tool frame will move from its initial location in a smooth manner until $\{T\} = \{G\}$ at the end of motion.

5. Examples

- ❑ Scenario: robotic arm grips a cup on the table and moves the arm to hang it on a cup holder on the wall.
- ❑ Current task: To enable the RRRRRR arm to grip the cup in the following posture (starting point C of the task), what are the 6 joint angles of the manipulator?

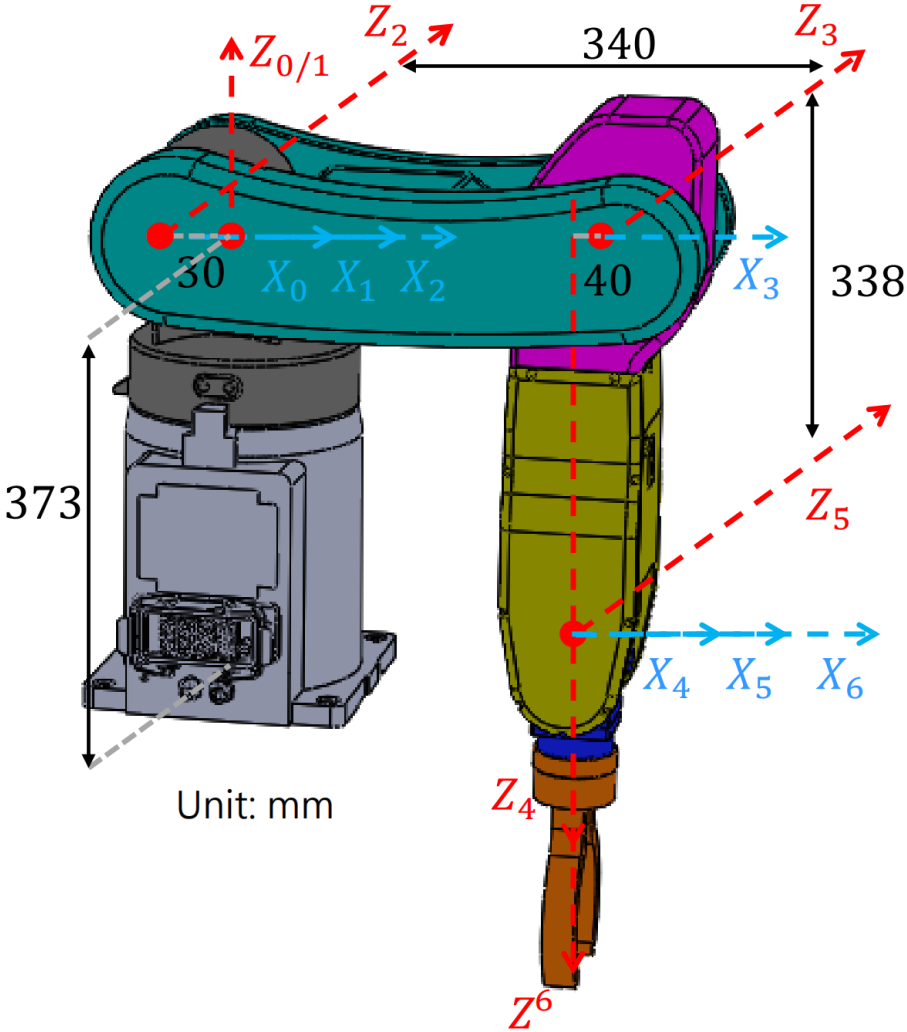


5. Examples

Step 1: Define DH Table

The graph shows the state of the axes at 0°

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0°	0	0	θ_1
2	-90°	$a_1 = -30$	0	θ_2
3	0°	$a_2 = 340$	0	θ_3
4	-90°	$a_3 = -40$	$d_4 = 338$	θ_4
5	90°	0	0	θ_5
6	-90°	0	0	θ_6

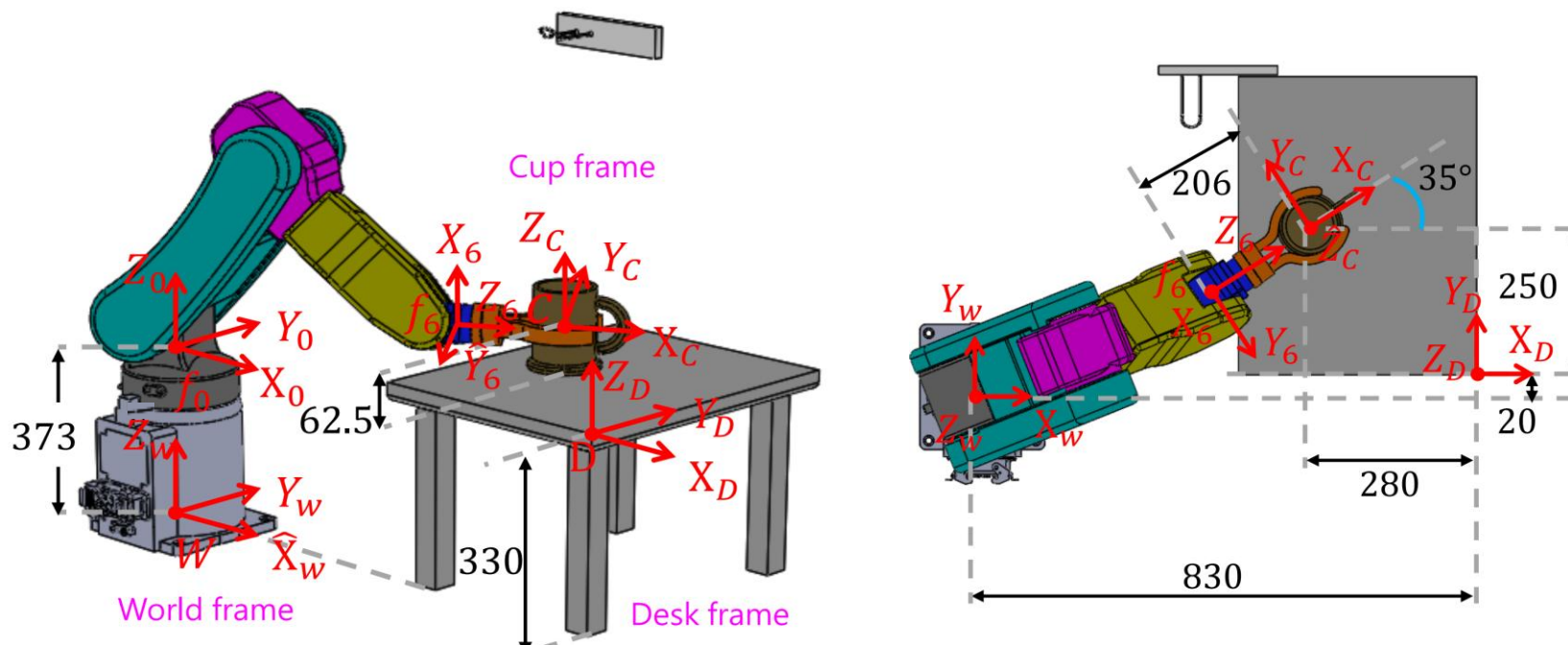


5. Examples

□ Step 2: Find ${}^W_C T$, and further find ${}^0_6 T$

$${}^W_C T = {}^W_D T {}^D_C T = \begin{bmatrix} 1 & 0 & 0 & 830 \\ 0 & 1 & 0 & 20 \\ 0 & 0 & 1 & 330 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos 35^\circ & -\sin 35^\circ & 0 & -280 \\ \sin 35^\circ & \cos 35^\circ & 0 & 250 \\ 0 & 0 & 1 & 62.5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Based on the relative relationship between “desk relative to manipulator” and “cup relative to desk”



5. Examples

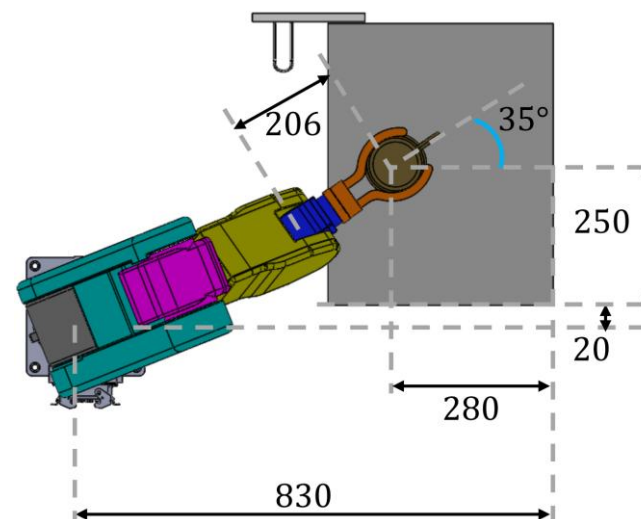
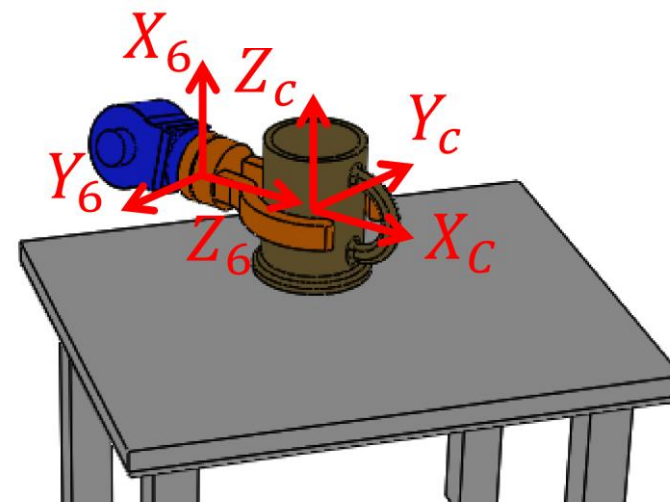
$${}^w_c T = {}^w_0 T {}^0_6 T {}^6_c T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 373 \\ 0 & 0 & 0 & 1 \end{bmatrix} {}^0_6 T \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 206 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0_6 T = {}^w_0 T^{-1} {}^w_c T^{-1}$$

$$= \begin{bmatrix} 0 & 0.5736 & 0.8192 & 381.3 \\ 0 & -0.8192 & 0.5736 & 151.8 \\ 1 & 0 & 0 & 19.5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0_6 R = \begin{bmatrix} 0 & 0.5736 & 0.8192 \\ 0 & -0.8192 & 0.5736 \\ 1 & 0 & 0 \end{bmatrix}$$

$${}^0 P_{6 ORG} = \begin{bmatrix} 381.3 \\ 151.8 \\ 19.5 \end{bmatrix}$$





5. Examples

□ Step 3: Find $\theta_1 - \theta_6$

- Solve $\theta_1, \theta_2, \theta_3$

$$\begin{bmatrix} f_1(\theta_3) \\ f_2(\theta_3) \\ f_3(\theta_3) \\ 1 \end{bmatrix} = {}^2_3T {}^3P_{4ORG} = \begin{bmatrix} c_3 & -s_3 & 0 & 340 \\ s_3 & c_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -40 \\ 338 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 340 - 338s_3 - 40c_3 \\ 338c_3 - 40s_3 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} g_1(\theta_2, \theta_3) \\ g_2(\theta_2, \theta_3) \\ g_3(\theta_2, \theta_3) \\ 1 \end{bmatrix} = {}^1_2T \begin{bmatrix} f_1(\theta_3) \\ f_2(\theta_3) \\ f_3(\theta_3) \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} c_2 & -s_2 & 0 & -30 \\ 0 & 0 & 1 & 0 \\ -s_2 & -c_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} f_1(\theta_3) \\ f_2(\theta_3) \\ f_3(\theta_3) \\ 1 \end{bmatrix} = \begin{bmatrix} 340c_2 - 40c_{23} - 338s_{23} - 30 \\ 0 \\ 40s_{23} - 338c_{23} - 340s_2 \\ 1 \end{bmatrix}$$



5. Examples

$$r = (k_1 c_2 + k_2 s_2) 2a_1 + k_3 = \|P\|^2 = 1.68498214e5$$

$$z = (k_1 s_2 - k_2 c_2) s \alpha_1 + k_4 = 19.5$$

Calculate $\theta_1, \theta_2, \theta_3$

$$\frac{(r - k_3)^2}{4a_1^2} + \frac{(z - k_4)^2}{s^2 \alpha_1} = k_1^2 + k_2^2$$

$$\Rightarrow \text{solve } \theta_3 = 2.5^\circ$$

$$r = (k_1 c_2 + k_2 s_2) 2a_1 + k_3$$

$$\Rightarrow \text{solve } \theta_2 = -52.2^\circ$$

$$x = c_1 g_1(\theta_2, \theta_3) - s_1 g_2(\theta_2, \theta_3)$$

$$\Rightarrow \text{solve } \theta_1 = 21.8^\circ$$



5. Examples

- Solve $\theta_4, \theta_5, \theta_6$

$${}^0_3R = \begin{bmatrix} 0.6006 & 0.7082 & -0.3710 \\ 0.24 & 0.2830 & 0.9286 \\ 0.7627 & -0.6468 & 0 \end{bmatrix}$$

$${}^3_6R = {}^0_3R^{-1} {}^0_6R = \begin{bmatrix} 0.7627 & 0.1477 & 0.6297 \\ -0.6468 & 0.1744 & 0.7424 \\ 0 & -0.9735 & 0.2286 \end{bmatrix}$$

Use the Z-Y-Z Euler angle to find the remaining joint angles

$$\theta_4 = -20^\circ \quad \theta_5 = -42^\circ \quad \theta_6 = 15^\circ$$

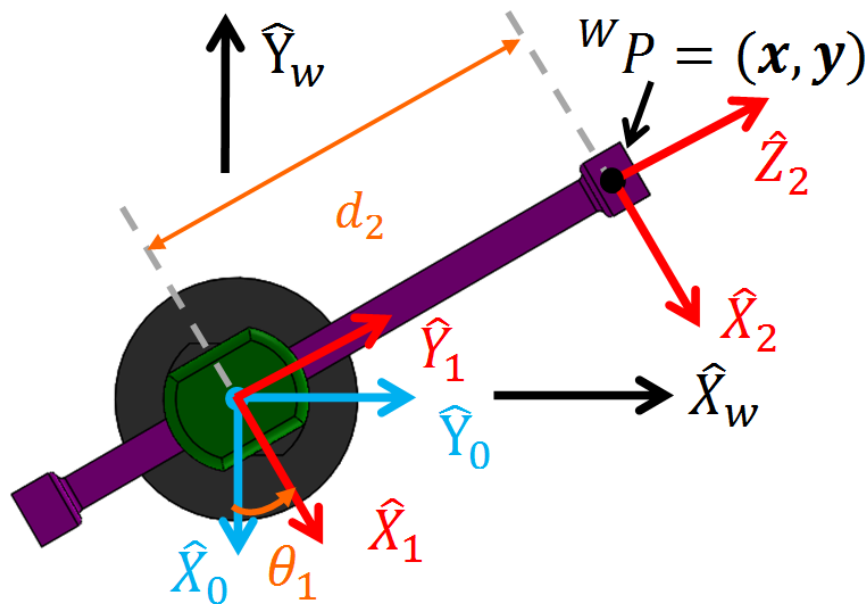


Summary

- ❑ The common methods for solving inverse kinematics.
- ❑ Pieper's solution to perform inverse kinematics calculations for 6-DOF robotic arms.
- ❑ Combine specific tasks to grasp the overall process of solving inverse kinematics, such as how to establish coordinate systems, calculate transformation matrices, etc.
- ❑ Think about the other inverse kinematics solving methods, such as neural networks, genetic algorithms, etc.

Assignment

- **Assignment 1:** Refer to the defined parameters of the RP robotic arm in the following figure, and develop a DH parameter table.



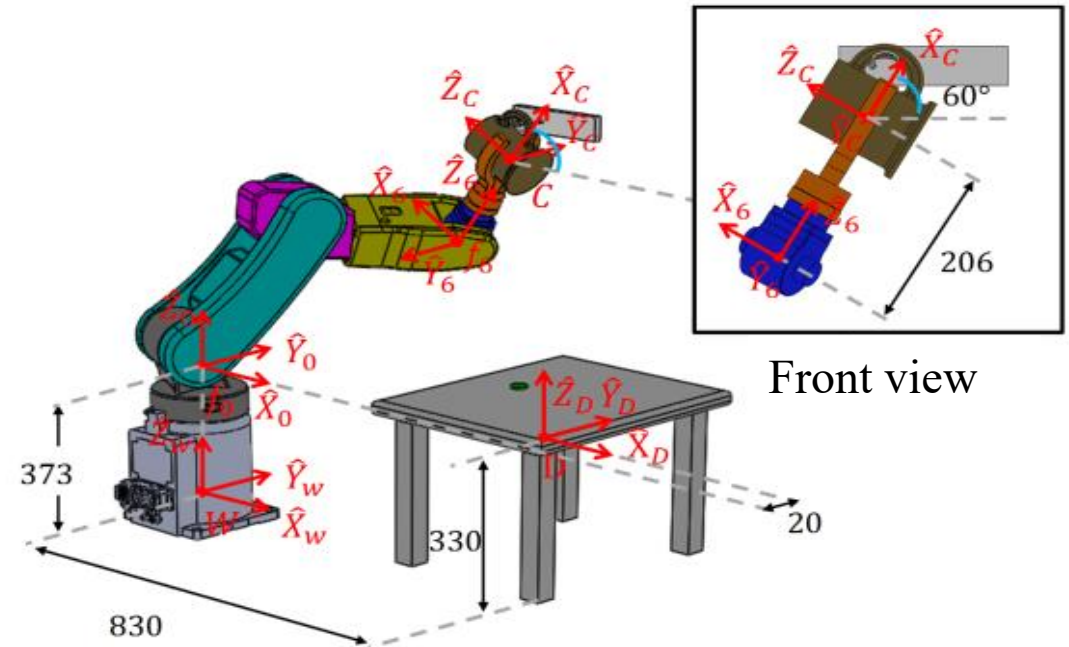
- **Assignment 2:** Continuing from the previous question, find the homogeneous transformation matrix 0_2T , represented by θ_1 and d_2 .



Assignment

□ **Assignment 3:** As shown in the figure on the next page, the diagram represents the robotic arm grasping a cup and placing it on the cup holder. The desired position and orientation of the end-effector are: ${}^D P_{C org} = [-500, 452, 410]^T$. The joint limits for each axis are: $\theta_1 = [-90, 90]$, $\theta_2 = [-180, 180]$, $\theta_3 = [-180, 180]$, $\theta_4 = [-90, 90]$, $\theta_5 = [-90, 90]$, $\theta_6 = [-90, 90]$. The Denavit-Hartenberg (DH) parameters are given in the table below. Find the homogeneous transformation matrix ${}^W_C T$.

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0°	0	0	θ_1
2	-90°	$a_1 = -30$	0	θ_2
3	0°	$a_2 = 340$	0	θ_3
4	-90°	$a_3 = -40$	$d_4 = 338$	θ_4
5	90°	0	0	θ_5
6	-90°	0	0	θ_6





Q&A